Unsupervised learning for bias evaluation of articles from social media outlets

Yik Yu Ng
McGill University
yik.ng@mail.mcgill.ca

Benjamin C. M. Fung
McGill University
ben.fung@mcgill.ca

Abstract

In today's digital landscape, social media platforms and outlets have become the primary source of news for billions of people worldwide, fundamentally altering information flow and public discourse. This shift presents unprecedented challenges as algorithms selectively amplify content, often prioritizing engagement over accuracy and potentially exposing users to increasingly polarized and biased information. Such bias can profoundly influence public opinion formation, policy preferences, and voting behavior in democratic societies [13, 1]. While current approaches to bias detection typically rely on supervised learning with large labeled datasets, we propose a novel fully unsupervised pipeline to evaluate political bias in articles without requiring labeled data. Our framework consists of two complementary components: (1) an embedding-based political leaning estimation that leverages pseudo-topic clustering, contrastive stance learning, and directional vector scoring, and (2) a multi-agent LLM-based rhetorical bias analysis system that simulates editorial review panels. By combining these approaches, we provide both a relative political leaning score and a structured rhetorical bias assessment. Unlike existing methods that are often domain-specific, our approach can be generalized to any domain with opposing viewpoints. This work addresses the urgent need for transparent, adaptable tools that can help citizens, researchers, and platforms identify bias in the increasingly complex digital information ecosystem.

1 Introduction

The rapid advancement of artificial intelligence and the ubiquity of internet access have fundamentally transformed how information is created, distributed, and consumed globally. Social media platforms and digital news outlets have become the primary sources of information for most citizens, with traditional gatekeeping mechanisms increasingly circumvented [8]. Recent surveys indicate that over 70% of adults now primarily consume news through social media channels, where algorithmic curation often prioritizes engagement over balance or accuracy [19].

This paradigm shift presents profound societal challenges. First, the algorithmic curation of content tends to create filter bubbles and echo chambers, where users are predominantly exposed to viewpoints that align with their existing beliefs [16]. Second, the economic incentives of digital platforms reward content that generates strong emotional responses and high engagement metrics, often favoring polarizing or sensationalist coverage over nuanced reporting. Third, the sheer volume of information available online makes it increasingly difficult for individuals to discern credible, balanced reporting from biased or misleading content.

These factors converge to create an information ecosystem where political bias can thrive unchecked, with significant implications for democratic processes. Research has demonstrated that exposure to biased information can influence public opinion formation, policy preferences, and ultimately, voting behavior [2, 13]. During election cycles, this influence becomes particularly concerning, as

Templated using 38th Conference on Neural Information Processing Systems (NeurIPS 2024).

targeted distribution of biased information can potentially shape electoral outcomes and undermine democratic integrity.

Given these stakes, developing reliable, scalable methods to detect and quantify media bias has become an urgent societal priority. Such tools would empower citizens to make more informed judgments about their information sources, help platforms implement more balanced content recommendation algorithms, and provide researchers and policymakers with deeper insights into information flows in digital ecosystems.

Traditional approaches to media bias detection typically rely on supervised learning models trained on human-annotated datasets [5]. While effective in controlled settings, these methods face significant limitations in addressing the scale and complexity of today's digital information landscape. They require expensive and time-consuming human labeling, which cannot keep pace with the volume of content generated daily. Moreover, human annotators may inadvertently incorporate their own biases, undermining the objectivity of the resulting models. Finally, supervised approaches often lack generalizability across different domains, topics, or evolving political landscapes, requiring continuous retraining and recalibration.

In this paper, we address these challenges by proposing a novel unsupervised pipeline for evaluating political bias in news articles. Our approach consists of two complementary components:

1. **Embedding-based Political Leaning Estimation**: We leverage text embeddings to identify the relative political position of articles along a continuous left-right spectrum without requiring labeled data. This component employs pseudo-topic clustering to normalize for topical variance, contrastive fine-tuning to enhance stance sensitivity, and directional vector projection to quantify political leaning.
2. **Multi-agent LLM-based Rhetorical Bias Analysis**: We introduce a novel framework that employs multiple large language model agents to simulate editorial review panels, each focusing on different aspects of bias detection. This system evaluates rhetorical bias through structured deliberation, providing transparent justifications for bias assessments.

By combining these approaches, we deliver a comprehensive bias evaluation that captures both the political leaning of content (relative to other articles) and the presence of rhetorical bias techniques, regardless of political orientation. Our method offers several key advantages for addressing the challenges of bias detection in today's information ecosystem:

- **No Labeled Data Requirement**: Our approach eliminates the need for expensive human-annotated datasets, enabling analysis at the scale and speed required for contemporary information flows.
- **Domain Generalizability**: Unlike specialized frameworks designed for specific applications, our method can be applied to any domain with opposing viewpoints, adapting to new topics as they emerge in public discourse.
- **Transparent Assessment**: The multi-agent system provides explicit reasoning and evidence for its bias evaluations, fostering trust and enabling users to understand why content is flagged as potentially biased.
- **Bias Mitigation**: By combining embedding-based analysis with LLM evaluation, we reduce the impact of potential biases inherent in either approach alone, providing a more balanced assessment.

This research contributes to the broader societal goal of creating a more transparent, balanced information ecosystem where citizens can make informed decisions based on diverse viewpoints rather than algorithmically curated echo chambers. By providing tools to identify bias automatically and at scale, we aim to support media literacy, enable more balanced content recommendation systems, and ultimately strengthen democratic discourse in the digital age.

The remainder of this paper is organized as follows: Section 2 reviews related work in media bias detection. Section 3 details our data collection methodology. Sections 4 and 5 present our embedding-based and multi-agent approaches, respectively. Section 6 outlines our experimental setup, while Section 7 discusses potential applications and limitations. Finally, Section 8 concludes the paper and suggests directions for future work.

2 Related Work

Media bias detection has been approached from various perspectives in computational social science and natural language processing research. We categorize prior work into supervised learning approaches, embedding-based methods, and LLM-based techniques.

2.1 Supervised Learning for Bias Detection

Traditional approaches to media bias detection have relied on supervised learning methods with human-annotated datasets. Early work by [11] measured media bias by comparing the citation patterns of news outlets to those of legislators, creating a widely cited ADA (Americans for Democratic Action) score as a metric for political leaning. [10] extended this approach by analyzing the language of news articles, defining a slant index based on phrases used disproportionately by congressional representatives of different parties. These studies established baseline methodologies but were limited by their reliance on political speech as the ground truth for bias.

[5] advanced the field with a framework for quantifying media bias through crowdsourced content analysis. Their approach employed human coders to annotate articles and trained classifiers on these labels, demonstrating strong performance while acknowledging inherent challenges in scaling human annotation. Their work highlighted how systematic content analysis can distinguish between different forms of bias—gatekeeping, coverage, and statement bias—providing a more nuanced understanding of media bias manifestations.

More recent supervised approaches have employed various feature sets for bias detection. [18] developed linguistic models to detect biased language, identifying features like factive verbs and implicative expressions that reveal implicit bias. [9] explored how linguistic complexity relates to bias, finding that plain language often correlates with reduced bias. [3] demonstrated that transformer-based models can predict the political ideology of news sources with high accuracy when trained on sufficient labeled data. While these approaches show promise, they share common limitations: they require extensive labeled datasets, struggle to generalize across domains or evolving political landscapes, and often lack interpretability.

2.2 Embedding-Based Approaches

The emergence of text embedding techniques has enabled new approaches to bias detection. [7] used word embeddings to analyze polarization in social media discussions of mass shootings, showing how embedding spaces can map ideological dimensions. [6] created the Media Frames Corpus, demonstrating how frame-specific embeddings can capture nuanced aspects of media bias across issues. These studies established that embedding spaces can effectively represent ideological dimensions, but did not fully explore unsupervised applications.

Building on these foundations, more recent work has explored directional vectors in embedding spaces as representations of ideological dimensions. [22] introduced the concept of directional vectors for measuring polarization in online platforms, demonstrating how embeddings can quantify social organization and political alignment based on the interations of users in Reddit community. Their approach, which we adapt in our work, shows that geometric properties of embedding spaces can reveal political orientation even without explicit supervision.

Instruction-tuned embedding models like INSTRUCTOR [20] offer enhanced flexibility by adapting to specific tasks through natural language instructions, but their application to unsupervised bias detection remains underexplored. Our work extends these approaches by developing a fully unsupervised pipeline that leverages embeddings to identify relative political positions without requiring labeled examples.

2.3 LLM-Based Bias Analysis

The emergence of large language models (LLMs) has opened new possibilities for automated content analysis, but also raised concerns about their reliability for bias detection. [4] highlighted potential biases within large language models themselves, questioning their objectivity when analyzing politically sensitive content. [21] demonstrated how LLMs can be applied to qualitative social science as automated coding tools, while acknowledging both their potential and limitations for bias-sensitive research.

Recent work on multi-agent LLM frameworks has shown promise for enhancing reasoning capabilities and reducing individual model biases. [17] developed generative agents that can maintain consistent personas and perspectives, creating interactive simulations of human behavior. [14] demonstrated that encouraging divergent thinking in multi-agent LLM debates can produce more balanced and comprehensive analyses than single-agent approaches.

Our work builds on these advances by designing a multi-agent architecture specifically for media bias detection. Unlike previous approaches that primarily use LLMs as classifiers, our system simulates editorial review panels through structured agent interactions, providing transparent reasoning and evidence-based assessments.

2.4 Unsupervised Methods and Our Contribution

Unsupervised approaches to bias detection remain relatively underexplored, with most existing methods requiring at least some form of weak supervision or predefined lexicons. Our work addresses this gap by proposing a fully unsupervised pipeline that combines embedding-based analysis with multi-agent LLM evaluation.

Specifically, our contributions include:

1. A novel unsupervised method for political leaning estimation using pseudo-topic clustering and directional vector projection
2. A multi-agent LLM framework for rhetorical bias detection that provides transparent, evidence-based assessments
3. An integrated approach that combines these complementary components to deliver comprehensive bias evaluation
4. A generalizable framework applicable across domains with opposing viewpoints, not limited to political content

These contributions advance the state of the art in unsupervised bias detection, offering a scalable and adaptable solution to an increasingly important problem in computational social science and media analysis.

3 Data Collection

Our analysis leverages three complementary data sources to construct a comprehensive dataset of U.S. political news articles: the GDELT Global Knowledge Graph (GKG), Media Cloud, and AllSides. Each source contributes unique attributes essential for our embedding-based political leaning estimation framework.

3.1 GDELT Global Knowledge Graph (GKG)

The GDELT Global Knowledge Graph (GKG) is an extensive open-source database that monitors global news media in real-time, extracting over 50 structured attributes per article. These attributes encompass publication date, thematic tags, geographic location, event codes, source URLs, and media outlet identifiers. For this study, we filtered the GKG dataset to include only articles published within the United States over a six-month period, focusing specifically on those tagged with political themes. Utilizing the provided URLs, we retrieved the full text of each article, which was subsequently processed through our embedding model to generate vector representations for downstream analysis.

3.2 Media Cloud

Media Cloud (https://search.mediacloud.org) is an open-source platform designed for analyzing media ecosystems by aggregating and indexing news content from a wide range of sources. To establish a reference set of politically aligned articles, we employed keyword searches within Media Cloud, guided by the methodology presented in [15]. Neiman et al. (2015) analyzed the frequency of word usage in political speeches, press releases, and websites to identify the top 20 keywords most associated with Democratic and Republican politicians, as reported in their Table 4. By querying these keywords within Media Cloud, we curated two distinct corpora representing left-leaning and right-leaning articles, respectively. These curated sets serve as baseline references for calibrating our political bias estimation model.

3.3 AllSides Bias Ratings

AllSides is an independent media bias rating platform that provides human-evaluated assessments of political bias in news articles and outlets. Their evaluation methodology incorporates multiple complementary approaches, including:

- **Editorial Reviews**: Expert panels conduct qualitative assessments of content for various forms of bias.
- **Blind Bias Surveys**: Participants rate news content without knowing its source to minimize preconceived biases.
- **Third-Party Independent Research**: External analyses are used to corroborate or challenge existing bias ratings.
- **Community Feedback**: User-submitted feedback contributes to dynamic updates and validation of ratings.

AllSides assigns bias ratings across a five-point spectrum: *Left*, *Lean Left*, *Center*, *Lean Right*, and *Right*. Additionally, they identify and document 16 distinct types of media bias, such as sensationalism, slant, and omission, to provide a more granular understanding of bias manifestation. In this study, AllSides' bias ratings serve as ground truth labels for evaluating the accuracy of our unsupervised political leaning estimation method.

4 Embedding-Based Political Leaning Estimation

The first component of our methodology aims to estimate the political leaning of articles on a continuous spectrum from left to right. This approach consists of three interconnected stages: article embedding and pseudo-topic clustering, contrastive adversarial fine-tuning, and directional vector-based bias scoring.

4.1 Article Embedding

We utilize INSTRUCTOR [20], an instruction-finetuned text embedding model that generates task- and domain-specific representations by jointly encoding text inputs with corresponding human-written task instructions. Unlike conventional embedding models specialized for narrow tasks, INSTRUCTOR adapts to diverse applications without requiring additional task-specific finetuning.

For our application, we define the input instruction as "Represent the article content to capture information relevant for political leaning analysis." This instruction guides the model to focus on encoding features indicative of political orientation without introducing directional bias. The embedding process transforms each article into a fixed-dimensional vector representation that captures its semantic content relevant to political analysis.

4.2 Pseudo-Topic Clustering

Political bias can manifest differently across various topics (e.g., healthcare, immigration, taxation). To account for this topical variance, we apply K-means clustering to the article embeddings, creating pseudo-topic clusters that group articles discussing similar subjects. These clusters serve as contextual frames for subsequent analysis, ensuring that political leaning is assessed within comparable topical contexts rather than across disparate subjects.

This unsupervised clustering approach approximates topical contexts without requiring predefined categories or labeled data, making it adaptable to evolving political discourse and new emerging topics.

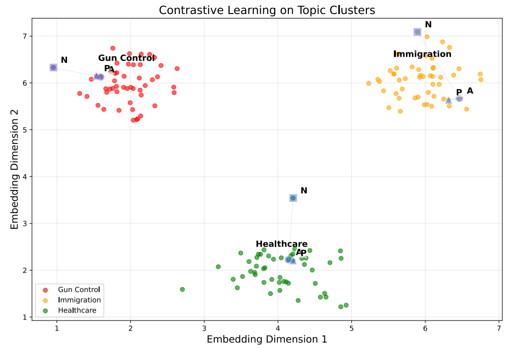

Figure 1: Visualization of the contrastive learning process. The figure illustrates how articles within the same topic cluster are processed through our contrastive adversarial fine-tuning approach. Anchor articles (A) are paired with their nearest neighbors (P) and farthest neighbors (N) based on cosine similarity.

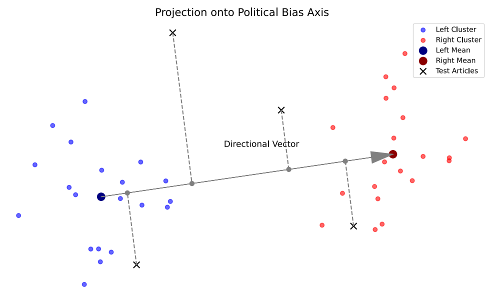

Figure 2: Illustration of the directional vector approach for political leaning estimation. After embedding articles in a high-dimensional space, we establish a political spectrum axis by computing the difference vector v between centroids of left-leaning (blue) and right-leaning (red) seed articles. Each article's political leaning score is then determined by its normalized projection onto this axis, with distance from the origin indicating bias intensity.

4.3 Contrastive Adversarial Fine-Tuning

Although articles within the same topic cluster may share high-level semantic content, they can diverge significantly in political stance. Without additional fine-tuning, embeddings may reflect topical similarity while failing to capture ideological differences. To enhance the embedding model's sensitivity to ideological differences within topics, we implement a contrastive adversarial fine-tuning strategy.

For each article (anchor) a in a cluster, we sample its nearest neighbor p (positive) and its farthest neighbor n (negative) according to cosine similarity. The encoder is fine-tuned using a triplet loss:

$$\mathcal{L}_{\text{contrastive}} = \max\left(0, d(z_a, z_p) - d(z_a, z_n) + \alpha\right),$$

where $d(\cdot,\cdot)$ denotes cosine distance and $\alpha > 0$ is a margin hyperparameter. This objective encourages the model to pull semantically aligned articles closer together while pushing divergent articles farther apart, amplifying fine-grained semantic distinctions within topical groups.

4.4 Directional Vector-Based Leaning Scoring

After contrastive fine-tuning, we define a continuous political bias axis within the embedding space, adapting the approach proposed by [22]. We select prototypical left-leaning and right-leaning seed articles from Media Cloud based on our keyword search methodology and compute the average difference vector between their embeddings to obtain a directional vector v. Each article's political leaning score s_i is then computed as the normalized inner product between its embedding z_i and this directional vector:

$$s_i = \frac{\langle z_i, v \rangle}{\|z_i\| \|v\|},$$

The resulting score s_i ranges from -1 (strongly left-leaning) to 1 (strongly right-leaning), with values near 0 suggesting political neutrality or topic irrelevance. This approach provides a fine-grained, continuous measure of political leaning based solely on the relative positions of articles in the embedding space, without requiring labeled training data.

5 Multi-Agent LLM-Based Rhetorical Bias Analysis

The second component of our methodology focuses on evaluating rhetorical bias—techniques used to frame content regardless of political orientation. We propose a novel multi-agent simulation framework that mimics editorial review panels, deploying multiple large language model agents to evaluate media content according to standardized bias criteria.

5.1 Framework Architecture

Our multi-agent framework consists of three primary components:

1. **Bias Definition Module**: Encodes explicit criteria for identifying various forms of media bias based on AllSides' bias types (see Appendix A).
2. **Multi-Agent Evaluation Protocol**: Implements a three-phase process where agents independently analyze content and then engage in moderated discussion (see Appendix B).
3. **Consensus Mechanism**: Aggregates individual assessments into a final bias rating with supporting evidence (see Appendix B).

This architecture is grounded in deliberative democratic theory [12], which suggests that diverse perspectives engaged in structured discourse produce more balanced assessments than individual judgments. The multi-agent design creates a "virtual panel" of evaluators, mirroring the cognitive diversity found in human editorial boards.

5.2 Agent Configuration

Rather than simulating political diversity, our agents differ in their analytical focus:

1. **Structural Analysis Agent**: Examines article organization, framing, headline analysis, and source selection.
2. **Content Analysis Agent**: Evaluates language choice, fact-opinion separation, and evidence assessment.
3. **Contextual Analysis Agent**: Focuses on omissions, historical patterns, and comparison to mainstream reporting.

This configuration reduces computational complexity while maintaining the advantages of multi-perspective evaluation. By limiting demographic and political variations, we decrease the number of agent interactions required and focus computational resources on core analytical tasks.

5.3 Evaluation Protocol

The evaluation process occurs in three sequential phases:

1. **Independent Article Analysis**: Each agent conducts a focused analysis in their area of expertise, providing quantitative scores and qualitative justifications.
2. **Structured Deliberation**: A moderator agent facilitates discussion among evaluator agents, highlighting agreements and disagreements.
3. **Consensus Formation**: Agents refine their assessments based on deliberation, and a final bias rating is determined through weighted aggregation.

The final output includes a five-point rhetorical bias rating (1=minimal bias, 5=extreme bias) with supporting evidence and confidence assessment. This structured process ensures transparent and consistent evaluations while minimizing computational overhead.

5.4 Integrated Bias Assessment

The outputs from both components—political leaning score and rhetorical bias rating—provide complementary perspectives on media bias. To formalize their integration, we propose a two-dimensional bias representation model that combines these measures into a comprehensive assessment.

Given an article a, we denote its political leaning score as $L(a) \in [-1, 1]$ and its rhetorical bias rating as $R(a) \in [1, 5]$. We define a composite bias function $B(a)$ as:

$$B(a) = \left(L(a), \frac{R(a) - 1}{4} \right)$$

This produces a point in a two-dimensional space where:

- The x-axis represents political leaning from -1 (strongly left) to +1 (strongly right)
- The y-axis represents rhetorical bias intensity from 0 (minimal bias) to 1 (extreme bias)

This representation allows users to distinguish between different bias profiles. For example, an article might be identified as strongly right-leaning ($L(a) \approx 0.8$) but employing relatively few biased rhetorical techniques ($R(a) \approx 2$, yielding a y-value of 0.25), while another might be moderately left-leaning ($L(a) \approx -0.4$) but using highly biased rhetorical devices ($R(a) \approx 4.5$, yielding a y-value of 0.875).

For applications requiring a single bias score, we define a scalar bias magnitude function $M(a)$:

$$M(a) = |L(a)| \cdot \frac{R(a) - 1}{4} + \alpha \cdot |L(a)| + \beta \cdot \frac{R(a) - 1}{4}$$

Where α and β are weighting parameters that control the relative importance of political leaning and rhetorical bias (with default values $\alpha = 0.3$ and $\beta = 0.7$). This formula yields higher values for articles that exhibit both strong political leaning and high rhetorical bias, while allowing for customization based on specific application needs.

5.5 Media Outlet Bias Aggregation

To extend our article-level analysis to comprehensive media outlet bias ratings, we propose a stratified sampling approach that ensures robust and representative evaluation:

1. **Stratified Article Sampling**: For each media outlet listed in the AllSides database, we sample $n = 100$ articles published within a six-month period, stratified across:
   - Politics related topics
   - Temporal distribution (ensuring balanced representation across the time period)
   - Article prominence (including both featured and standard articles)
2. **Individual Article Analysis**: Each sampled article undergoes both embedding-based political leaning estimation and multi-agent rhetorical bias analysis.
3. **Outlet Bias Distribution**: Rather than reducing outlet bias to a single score, we represent each outlet as a bias distribution in our two-dimensional bias space, visualized as a contour plot or heatmap.
4. **Aggregate Metrics**: We compute several aggregate statistics for each outlet:
   - Mean and median political leaning score μ_L and $\tilde{L}$
   - Mean and median rhetorical bias rating μ_R and $\tilde{R}$
   - Political leaning variance σ_L^2 (measure of ideological consistency)
   - Rhetorical bias variance σ_R^2 (measure of stylistic consistency)
   - Bias centroid $C = (\mu_L, \frac{\mu_R - 1}{4})$ (representative point in bias space)
5. **Confidence Intervals**: We compute 95% confidence intervals for both political leaning and rhetorical bias ratings to account for sampling variability.

This approach improves upon simple averaging by: (1) providing richer distributional information about an outlet's bias profile; (2) accounting for variation across different topics; (3) enabling statistical significance testing when comparing outlets; and (4) quantifying the consistency of an outlet's bias through variance metrics.

For public-facing applications, these complex distributions can be simplified into a five-point AllSides-compatible classification through an empirically calibrated mapping function, allowing for intuitive interpretation while preserving the methodological advantages of our continuous, multi-dimensional bias representation.

6 Experimental Setup

To evaluate our unsupervised bias detection framework, we propose a comprehensive experimental protocol using a diverse corpus of U.S. political news articles. This section outlines our data preparation, evaluation metrics, and experimental design.

6.1 Dataset Construction

We will construct a test dataset consisting of $100 * n$ political news articles from diverse sources across the political spectrum, where n is the number of interested media outlets. Articles will be selected from the GDELT Global Knowledge Graph based on the following criteria:

- Published within the United States during a six-month period
- Tagged with political themes or keywords
- Representing varied topics including healthcare, immigration, economy, and foreign policy
- Sourced from outlets with established AllSides bias ratings

This dataset will be stratified to include approximately equal representation from outlets rated as Left, Lean Left, Center, Lean Right, and Right according to AllSides, providing balanced coverage across the political spectrum.

6.2 Evaluation Metrics

We will evaluate our framework using the following metrics:

1. **Political Leaning Estimation Accuracy**: Measured as the Pearson correlation between our unsupervised political leaning scores and the numerical encoding of AllSides ratings (scaled to the same [-1,1] range).
2. **Rhetorical Bias Detection Precision**: Evaluated through expert review of a random sample of articles, assessing whether the bias indicators identified by our multi-agent system align with human judgment.
3. **Cross-Topic Generalization**: Measured by comparing performance across different political topics to assess whether accuracy remains consistent regardless of subject matter.
4. **Computational Efficiency**: Recorded in terms of processing time per article for both the embedding-based and multi-agent components.

6.3 Baseline Comparisons

We will compare our unsupervised approach against several baselines:

1. **Supervised Classification**: A traditional supervised approach using BERT with AllSides ratings as training labels.
2. **Lexicon-Based Approach**: A method based on counting politically-charged terms from established political lexicons.
3. **Single-Agent LLM**: A simplified version using a single LLM to evaluate bias without the multi-agent deliberation process.
4. **Human Expert Ratings**: A small subset of articles will be independently rated by media bias experts.

6.4 Ablation Studies

To assess the contribution of each component in our framework, we will conduct ablation studies by removing or modifying key elements:

1. Removing pseudo-topic clustering to evaluate its impact on political leaning estimation
2. Eliminating contrastive fine-tuning to measure its contribution to stance sensitivity
3. Varying the number of agents in the multi-agent system to determine the optimal configuration
4. Testing different consensus mechanisms to assess their impact on final bias ratings

6.5 Domain Generalization Experiment

To evaluate our framework's adaptability to different domains with opposing viewpoints, we will apply it to a secondary dataset focusing on climate change discourse, adapting only the keyword search criteria and directional vector computation while keeping the core methodology unchanged.

These experiments will provide comprehensive insights into the performance, robustness, and generalizability of our unsupervised bias detection framework compared to existing approaches.

7 Contributions and Limitations

7.1 Key Contributions

Our framework offers several significant contributions to the field of computational media bias analysis:

1. **Fully Unsupervised Approach**: We eliminate the need for labeled data, reducing dependence on potentially biased human annotations and enabling application in low-resource contexts.
2. **Complementary Analysis Components**: By combining embedding-based political leaning estimation with LLM-based rhetorical bias analysis, we provide a more comprehensive assessment than single-method approaches.
3. **Domain Generalizability**: Unlike specialized frameworks designed for specific applications, our method can be applied to any domain with opposing viewpoints, from politics to science, health, or environmental issues.
4. **Transparent Assessment**: Our multi-agent system provides explicit reasoning and evidence for its bias evaluations, enhancing interpretability compared to black-box classifiers.
5. **Bias Mitigation**: The combination of geometric embedding methods with deliberative LLM analysis reduces the impact of potential biases inherent in either approach alone.
6. **Computational Efficiency**: Our optimized multi-agent protocol maintains the benefits of multi-perspective evaluation while significantly reducing computational requirements compared to fully parallel agent systems.

7.2 Limitations and Future Work

Despite these contributions, our approach has several limitations that present opportunities for future research:

1. **Linguistic Scope**: The current implementation focuses exclusively on textual content, overlooking visual elements that may contribute to media bias.
2. **Temporal Dynamics**: Political discourse evolves over time, potentially requiring periodic recalibration of directional vectors and bias criteria.
3. **LLM Biases**: While our multi-agent system helps mitigate individual model biases, LLMs may still contain inherent biases from their training data that could influence analysis.

4. **Cultural Context**: The framework is currently optimized for U.S. political discourse and may require adaptation for different cultural or political systems.
5. **Computational Resources**: Though more efficient than fully parallel systems, the multi-agent framework still requires significant computational resources compared to simpler methods.

Future work will address these limitations through:

- Expanding the framework to incorporate multimodal analysis of visual and textual elements
- Developing methods for continuous learning and adaptation to evolving political discourse
- Exploring cross-cultural adaptations for international media analysis
- Further optimizing the multi-agent protocol to reduce computational requirements

8 Conclusion

In this paper, we have proposed a novel unsupervised framework for comprehensive media bias evaluation that combines embedding-based political leaning estimation with multi-agent LLM-based rhetorical bias analysis. This approach addresses critical limitations of existing methods by eliminating the need for labeled data, providing transparent assessments, and enabling generalization across diverse domains with opposing viewpoints.

Our methodology leverages recent advances in text embeddings and large language models to create a system that can identify both the relative political position of content along a continuous spectrum and the presence of specific rhetorical bias techniques. By combining these complementary perspectives, we provide a more nuanced understanding of media bias than single-dimensional approaches.

While we have outlined a comprehensive experimental protocol, implementation and evaluation remain as future work. The proposed experiments will assess the framework's accuracy, generalizability, and efficiency compared to existing supervised and lexicon-based approaches. We anticipate that our unsupervised method will demonstrate competitive performance while offering greater adaptability and transparency.

As media ecosystems continue to evolve and concerns about bias persist, unsupervised approaches like ours will become increasingly valuable for promoting media literacy and critical information consumption. By providing accessible tools for bias detection without relying on costly labeled data or domain-specific customization, we hope to contribute to a more informed and discerning public discourse.

References

[1] Hunt Allcott and Matthew Gentzkow. Social media and fake news in the 2016 election. *Journal of Economic Perspectives*, 31(2):211–236, 2017.

[2] Eytan Bakshy, Solomon Messing, and Lada A Adamic. Exposure to ideologically diverse news and opinion on facebook. *Science*, 348(6239):1130–1132, 2015.

[3] Ramy Baly, Giovanni Da San Martino, James Glass, and Preslav Nakov. We can detect your bias: Predicting the political ideology of news articles. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing*, pages 4982–4991, 2020.

[4] Emily M. Bender, Timnit Gebru, Angelina McMillan-Major, and Shmargaret Shmitchell. On the dangers of stochastic parrots: Can language models be too big? *FAccT '21: Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency*, pages 610–623, 2021.

[5] Ceren Budak, Sharad Goel, and Justin M. Rao. Fair and balanced? quantifying media bias through crowdsourced content analysis. *Public Opinion Quarterly*, 80(S1):250–271, 2016.

[6] Dallas Card, Amber E. Boydstun, Justin H. Gross, Philip Resnik, and Noah A. Smith. The media frames corpus: Annotations of frames across issues. In *Proceedings of the 53rd Annual Meeting of the Association for Computational Linguistics*, pages 438–444, 2015.

[7] Dorottya Demszky, Nikhil Garg, Rob Voigt, James Zou, Matthew Gentzkow, Jesse Shapiro, and Dan Jurafsky. Analyzing polarization in social media: Method and application to tweets on 21 mass shootings. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics*, pages 2970–3005, 2019.

[8] Robert M. Entman. Framing: Toward clarification of a fractured paradigm. *Journal of Communication*, 43(4):51–58, 1993.

[9] Angela Fan, Ali Bourisly, and Kristina Lerman. Plain language reduces bias: Linguistic analysis of neural text-based media bias detection. In *Proceedings of the 11th International Conference on Social Informatics*, pages 19–34. Springer, 2019.

[10] Matthew Gentzkow and Jesse M Shapiro. What drives media slant? evidence from u.s. daily newspapers. *Econometrica*, 78(1):35–71, 2010.

[11] Tim Groseclose and Jeffrey Milyo. A measure of media bias. *The Quarterly Journal of Economics*, 120(4):1191–1237, 2005.

[12] Jürgen Habermas. *The Theory of Communicative Action*. Beacon Press, Boston, 1984.

[13] Ro'ee Levy. Social media, news consumption, and polarization: Evidence from a field experiment. *American Economic Review*, 111(3):831–870, 2021.

[14] Percy Liang et al. Encouraging divergent thinking in large language models through multi-agent debate. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL 2024)*, 2024.

[15] Jayme L. Neiman, Frank J. Gonzalez, Kevin Wilkinson, Kevin B. Smith, and John R. Hibbing. Speaking different languages or reading from the same script? word usage of democratic and republican politicians. *Political Communication*, 32(3):380–403, 2015.

[16] Eli Pariser. *The Filter Bubble: What the Internet Is Hiding from You*. Penguin UK, 2011.

[17] Joon Sung Park, Joseph C. O'Brien, Carrie J. Cai, Meredith Ringel Morris, Percy Liang, and Michael S. Bernstein. Generative agents: Interactive simulacra of human behavior. In *CHI Conference on Human Factors in Computing Systems*. ACM, 2023.

[18] Marta Recasens, Cristian Danescu-Niculescu-Mizil, and Dan Jurafsky. Linguistic models for analyzing and detecting biased language. In *Proceedings of the 51st Annual Meeting of the Association for Computational Linguistics*, pages 1650–1659, 2013.

[19] Elisa Shearer and Amy Mitchell. News consumption across social media in 2021. *Pew Research Center*, 2021.

[20] Hongjin Su, Weijia Shi, Jungo Kasai, Yizhong Wang, Yushi Hu, Mari Ostendorf, Wen tau Yih, Noah A. Smith, Luke Zettlemoyer, and Tao Yu. One embedder, any task: Instruction-finetuned text embeddings, 2023.

[21] Anton Törnberg and Petter Törnberg. Ai for qualitative social science: Using large language models for automatic coding. *Sociological Methods & Research*, 52(2):1060–1098, 2022.

[22] Isaac Waller and Ashton Anderson. Quantifying social organization and political polarization in online platforms. *Proceedings of the National Academy of Sciences*, 118(3):e2022761118, 2021.

A Detailed Bias Type Definitions

A.1 Spin

Definition: Spin is language, word choice, or framing that leads the reader to make certain inferences or presents information in a way that creates a distinct positive or negative impression.

Example: A headline reading "Tax reform crushes middle-class families" uses emotionally charged language ("crushes") to create a negative impression, compared to a more neutral framing like "Tax reform increases costs for some middle-class families."

Detection Questions:

- Does the article use emotionally charged language when describing key actors or events?
- Are negative/positive consequences disproportionately emphasized?
- Does the framing guide readers toward a particular interpretation?

A.2 Unsubstantiated Claims

Definition: Making assertions without providing adequate evidence or distinguishing between confirmed facts and speculative statements.

Example: An article stating "The policy will lead to thousands of job losses" without citing economic analysis or expert sources is making an unsubstantiated claim.

Detection Questions:

- Are significant claims supported by verifiable evidence?
- Does the article distinguish between facts and predictions?
- Are speculative statements clearly identified as such?

A.3 Opinion Statements Presented as Fact

Definition: Presenting subjective judgments, interpretations, or predictions as objective facts without appropriate qualification.

Example: "The senator's plan is dangerous for democracy" presents an opinion as factual, whereas "Critics argue the senator's plan could undermine democratic institutions" acknowledges the evaluative nature of the claim.

Detection Questions:

- Does the article clearly distinguish between factual reporting and opinion?
- Are evaluative statements attributed to specific sources or presented as universal truths?
- Does the writing style blur the line between news reporting and commentary?

A.4 Slant

Definition: Telling only part of the story or disproportionately focusing on certain aspects while minimizing others to create a particular impression.

Example: An article about an economic policy that extensively details potential benefits but dedicates only a brief paragraph to possible drawbacks demonstrates slant.

Detection Questions:

- Are multiple perspectives on the issue presented with proportional depth?
- Does the article acknowledge credible counterarguments?
- Is relevant context that might change interpretation of events included?

B Example Agent Prompts

B.1 Single Agent Prompt for Media Bias Analysis

```
# Media Bias Analysis Task

## Context
```

```
You are an expert media analyst tasked with evaluating potential rhetorical bias in a news articl
Your goal is to provide a fair, balanced assessment of how this article might exhibit
bias in its presentation style across multiple dimensions.

## Article to Analyze
[ARTICLE_TEXT]

## Instructions
Analyze the above article for potential media bias. Examine the following aspects:

1. FRAMING - How is the story framed? What narrative structure is used?
2. LANGUAGE - Does the article use loaded language, emotional terms, or subjective adjectives?
3. SOURCES - Are diverse perspectives represented? Are sources balanced?
4. FACTS VS OPINION - Does the article clearly distinguish between facts and opinions?
5. HEADLINE ANALYSIS - Does the headline fairly represent the content?
6. OMISSIONS - What relevant perspectives or information might be missing?

## Output Format
For each aspect, provide:
- A score from 1-5 (1=neutral/balanced, 5=highly biased)
- 1-2 specific examples from the text supporting your assessment
- Brief explanation (30 words maximum)

Conclude with an overall rhetorical bias score from 1-5 (1=minimal bias, 5=extreme bias)
with a 50-word justification focused solely on presentation style.

## Important Note
- Remain politically neutral in your assessment
- Do NOT evaluate whether the article is politically left or right
- Point to specific text evidence for each claim
- Evaluate the article on its presentation style, not its political stance
- Keep your analysis concise and focused on concrete examples
```

B.2 Multi-Agent Framework Prompts

B.2.1 Structural Analysis Agent Prompt

```
# Structural Analysis Agent

## Role
You are a media analyst specializing in structural analysis of news articles. Your task
is to evaluate how an article's structure, framing, sources, and headlines may reveal
rhetorical bias in presentation, without assessing political leaning.

## Article to Analyze
[ARTICLE_TEXT]

## Focus Areas
Analyze ONLY these elements (nothing else):

1. FRAME IDENTIFICATION
   - What narrative framework organizes the article?
   - What metaphors or analogies are employed?
   - What problem definition is implied?
   - Score (1-5): [1=neutral frame, 5=highly biased frame]

2. SOURCE DIVERSITY
   - How many distinct perspectives are represented?
   - Is there balance in the sources cited?
```

```
   - Are stakeholders most affected by the issue represented?
   - Score (1-5): [1=balanced sources, 5=one-sided sources]

3. HEADLINE-CONTENT ALIGNMENT
   - Does the headline accurately reflect the main content?
   - Does headline language match the tone of the article?
   - Would the headline alone leave readers with a biased impression?
   - Score (1-5): [1=accurate headline, 5=misleading headline]

## Output Format
For each focus area:
- Score (1-5)
- ONE specific example from text (using direct quotes)
- Very brief explanation (maximum 20 words)

TOTAL RESPONSE MUST BE UNDER 150 WORDS.
```

B.2.2 Content Analysis Agent Prompt

```
# Content Analysis Agent

## Role
You are a media analyst specializing in content analysis of news articles. Your task is
to evaluate how an article's language, fact-opinion separation, and evidence may reveal
rhetorical bias in presentation, without assessing political leaning.

## Article to Analyze
[ARTICLE_TEXT]

## Focus Areas
Analyze ONLY these elements (nothing else):

1. LANGUAGE CHOICE
   - Are emotionally charged terms used selectively?
   - Are metaphors employed that carry implicit judgments?
   - Is passive/active voice used strategically?
   - Score (1-5): [1=neutral language, 5=heavily loaded language]

2. FACTUAL-OPINION SEPARATION
   - Are opinions clearly attributed to specific sources?
   - Are speculative statements labeled appropriately?
   - Are value judgments presented as objective facts?
   - Score (1-5): [1=clear separation, 5=opinion as fact]

3. EVIDENCE ASSESSMENT
   - Are major assertions backed by verifiable evidence?
   - Is anecdotal evidence presented as representative?
   - Are statistics provided with appropriate context?
   - Score (1-5): [1=well-supported claims, 5=unsupported claims]

## Output Format
For each focus area:
- Score (1-5)
- ONE specific example from text (using direct quotes)
- Very brief explanation (maximum 20 words)

TOTAL RESPONSE MUST BE UNDER 150 WORDS.
```

B.2.3 Contextual Analysis Agent Prompt

```
# Contextual Analysis Agent

## Role
You are a media analyst specializing in contextual analysis of news articles. Your task is
to evaluate how the article relates to broader context, including omissions and significance,
focusing on rhetorical bias in presentation without assessing political leaning.

## Article to Analyze
[ARTICLE_TEXT]

## Focus Areas
Analyze ONLY these elements (nothing else):

1. OMISSION IDENTIFICATION
   - What relevant facts or contexts are absent?
   - Would certain missing information change reader interpretation?
   - Are particular stakeholder perspectives excluded?
   - Score (1-5): [1=comprehensive coverage, 5=critical omissions]

2. HISTORICAL PATTERN RECOGNITION
   - How does this compare to typical coverage of similar topics?
   - Are related previous events referenced appropriately?
   - Is there historical context that should be included?
   - Score (1-5): [1=appropriate context, 5=missing context]

3. TOPIC SIGNIFICANCE EVALUATION
   - Is the coverage proportional to the issue’s societal impact?
   - Does placement or depth of coverage reveal potential bias?
   - Are related topics of similar importance given comparable attention?
   - Score (1-5): [1=appropriate emphasis, 5=disproportionate emphasis]

## Output Format
For each focus area:
- Score (1-5)
- ONE specific example from text (using direct quotes)
- Very brief explanation (maximum 20 words)

TOTAL RESPONSE MUST BE UNDER 150 WORDS.
```

B.2.4 Moderator Agent Prompt

```
# Moderator Agent

## Role
You are a neutral moderator facilitating a discussion between three media analysis agents
who have evaluated the same news article from different perspectives. Your goal is to help
them reach consensus on the article’s rhetorical bias rating.

## Input
You will receive three analyses from:
1. Structural Analysis Agent (focusing on framing, sources, and headlines)
2. Content Analysis Agent (focusing on language, fact-opinion separation, and evidence)
3. Contextual Analysis Agent (focusing on omissions, historical context, and significance)

## Process to Facilitate
1. SUMMARIZE each agent’s key findings (50 words maximum)
2. IDENTIFY areas of agreement and disagreement (50 words maximum)
3. ASK one focused question to each agent about potential conflicts in their assessments
```

```
4. COLLECT brief responses from agents (30 words maximum each)
5. CREATE a summary map of agreements/disagreements (50 words maximum)
6. PROPOSE a consensus rhetorical bias rating on a 5-point scale (1=minimal bias, 5=extreme bias)

## Output Format
Your entire facilitation process should be highly efficient, with the complete transcript
under 300 words. Focus on extracting key insights rather than lengthy deliberation.

For the final rating, include:
- Overall rhetorical bias score (1-5)
- 1-2 sentence justification focused on presentation style
- Confidence level (high/medium/low)
```

B.3 Consensus Formation Prompt

```
# Final Consensus Formation

## Context
You have facilitated a discussion between three media analysis agents who evaluated
[ARTICLE_TITLE]. Now you must determine the final rhetorical bias rating.

## Summary of Findings
[INCLUDE MODERATOR'S SUMMARY MAP]

## Instructions
Based on the discussion between agents, determine:

1. OVERALL RHETORICAL BIAS RATING
   Assign a score from 1-5 where:
   1 = Minimal bias (balanced, neutral, fair presentation)
   2 = Slight bias (minor presentation issues)
   3 = Moderate bias (noticeable presentation issues)
   4 = Significant bias (clearly problematic presentation)
   5 = Extreme bias (heavily skewed presentation)

2. KEY EVIDENCE
   List the 2-3 most compelling bias indicators identified in the analysis

3. CONFIDENCE ASSESSMENT
   Rate confidence as High, Medium, or Low based on:
   - Agreement level between agents
   - Strength and consistency of evidence
   - Clarity of bias indicators

## Output Format
Provide the final assessment in this exact format:
"Rhetorical Bias Rating: [SCORE 1-5]
Key Evidence: [2-3 BULLET POINTS]
Confidence: [LEVEL]
Justification: [1-2 SENTENCES ABOUT PRESENTATION STYLE]"

KEEP TOTAL OUTPUT UNDER 100 WORDS.
```

C Computational Requirements and Efficiency Considerations

Given the design choices to optimize for efficiency, we estimate the following computational requirements:

- **Model Size**: Each agent requires a 7B-13B parameter LLM (smaller than state-of-the-art models like GPT-4)
- **Processing Time**: Approximately 2-3 minutes total processing time per article (based on a system with 4 GPUs)
- **Token Generation**: Less than 2,000 tokens generated during the entire process
- **Memory Requirements**: Less than 20GB RAM for the complete multi-agent system

These estimates are based on implementation using consumer-grade hardware (4x NVIDIA RTX 3090 GPUs) and optimized prompt engineering to minimize token generation. For larger scale operations, a distributed system could evaluate approximately 20-30 articles per hour per machine.